

REQUEST FOR INFORMATION

United States Marine Corps Advanced Amphibious Assault Program

NOTE: THIS IS NOT A SOLICITATION FOR PROPOSALS. THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is for planning and informational purposes only and is not a Request for Proposal (RFP) and is not an obligation on the part of the Government to acquire any products or services. The Government will NOT be responsible for any costs incurred in responding to this RFI or furnishing the requested information. No contract will be awarded from this announcement. Pursuant to Federal Acquisition Regulations (FAR) 15.201(e), responses to this RFI are not offers and cannot be accepted by the Government to form a binding contract.

1.0 Description

1.1 Background

Emerging threats on today's battlefield require investigation and assessment of potential and rapidly available solutions to these threats. Given the lineup of 30mm x 173 ("30mm") based weapon systems fielded or in development, one such identified solution is the airburst capable 30mm cartridge or round. The capabilities of different 30mm airburst rounds have been demonstrated against a wide range of targets in past test programs. However, capabilities and configurations have evolved, requiring a reassessment of these available solutions.

The United States Marine Corps (USMC) is currently working to field an Amphibious Combat Vehicle (ACV) Mission Role Variant (MRV) armed with a 30mm Remote Weapon Station (RWS) manufactured by Kongsberg Defence & Aerospace (KDA). This MRV, identified as the ACV-30, employs a 30mm Mark (MK) 44 Modification (MOD) 4 Automatic Gun as its primary weapon. The MK 44 MOD 4 is a member of the Bushmaster® II Chain Gun® family. The USMC is investigating the potential integration of a 30mm airburst solution into the ACV-30 to address both current and emerging threat sets.

1.2 Purpose

This market research inquiry seeks technical data and other related information on 30mm airburst rounds currently produced, previously produced, or in active development by Respondent. For the purposes of this inquiry, an airburst round is one that contains at least one deliberate mode of operation that initiates or otherwise activates the warhead or payload while in flight prior to impact.

In addition, this inquiry also seeks technical data and program information for the combat vehicle or ship systems that have integrated or demonstrated any of these capabilities. The collected data will facilitate United States Government (USG) evaluation of airburst munition sources, capabilities, qualification status, integration, and similar follow-on activities.

1.3 Data Rights

It is acknowledged that some or all of the requested data may be considered Proprietary in nature to the Respondent or its partner(s) and that such data is provided in confidence to the USG; namely, the Department of the Navy (DoN). To the extent that data provided by the Respondent is clearly and appropriately marked as Proprietary, the Government will appropriate protect such information applicable statutory and Department of Defense (DoD) policy and instruction.

1.4 Classification

This inquiry is Unclassified by design. Unless specifically declared otherwise by the Respondent, any response received will be treated as Unclassified or Controlled Unclassified Information depending on the source markings and declarations of the Respondent. Information deemed Classified will be handled in the appropriate fashion as directed by DoD policy.

1.5 ACV-30 Overview

- 1.5.1 The Program Manager, Advanced Amphibious Assault (PM AAA), Program Executive Officer Land Systems (PEO LS), manages the ACV Family of Vehicles (FoV), to include the ACV-30. The ACV is a fully amphibious armored personnel carrier capable of ship-to-objective maneuvering across austere land-based environments. PM AAA intends to develop, integrate, and certify a programmable air burst munition (PABM) compatible with the ACV-30. The ACV-30 uses a RWS fitted with a MK44 Mod 4 automatic gun and a dual linkless feed ammunition system.
- 1.5.2 The ACV-30 is designed as an ACV-Personnel (ACV-P) lethality upgrade and provides more lethal fires at ranges greater than the ACV-P RWS. It provides a broader range of anti-personnel and anti-armor ammunition types with accurate fires while afloat, on the move ashore, or stationary from a battle position. The ACV-30 also has an improved capacity to defeat threats to the vehicle and growth capacity to meet emerging demands for counter unmanned aerial systems.
- 1.5.3 The ACV-30, as part of the ACV FoV, is a fully amphibious-capable vehicle. Its operating profile covers both land and waterborne environments. The system can launch and recover from naval shipping and land ashore while carrying embarked infantry to secure objectives. Once ashore, it can traverse hundreds of miles inland across difficult terrain. These vehicles consistently operate in maritime environments and face significant exposure to salt water and fog.

1.6 30mm RWS

- 1.6.1 The ACV-30 employs KDA's PROTECTOR Series Remote Turret – 20 (RT-20). The RT-20 is a gyro-stabilized design for remote operation. The operator can remotely control the traverse, elevation, and firing operations. It is equipped with advanced day and thermal imaging sensors, an eye-safe laser range finder, and an automatic weather station.

- 1.6.2 The RT-20 carries an ammunition handling system that supports a dual linkless feed, closed bolt, externally powered, chain driven, and PABM capable 30mm x 173 MK44 Mod 4 automatic gun.

1.7 Fire Control System

- 1.7.1 The ACV-30 fire control system was developed by KDA and allows the operator to observe, select, and engage moving targets with high precision. The fire control system leverages the azimuth and elevation motors of the sights and turret for a four-axis solution to continuously and precisely apply ballistic offset while keeping the target centered in the display at all times.

2.0 Applicable Documents

Certain USG DoD military standards (MIL-STD) and North Atlantic Treaty Organization (NATO) Standardization Agreements (STANAGs) are referenced in this inquiry. Table 1 provides a summary list of these documents. Please note that Table 1 is not a complete list of the applicable standards, instructions, specifications, agreements, and other requirements that will apply during any USG ammunition effort addressing airburst munitions produced or developed by the Respondent.

Table 1. Summary of Documents Referenced in the Market Research Inquiry.

NUMBER	TITLE	REVISION	DATE
MIL-STD-331	Fuze and Fuze Components, Environment and Performance Tests For	D	31 May 2017
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment	G	11 December 2015
MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems	D	24 December 2020
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests	H, Change 1	18 May 2022
MIL-STD-882	System Safety	E, Change 1	27 September 2023
MIL-STD-1316	Safety Criteria for Fuze Design	F	18 August 2017
MIL-STD-2105	Hazard Assessment Tests for Non-Nuclear Munitions	E	6 January 2022
MIL-STD-3029	Hot Gun Cook-Off Hazards Assessment, Test and Analysis	-	23 July 2009
NAVSEAINST 8020.5	Qualification and Final (Type) Qualification Procedures for Navy Explosives (High Explosives, Propellants, and Pyrotechnics)	D	1 April 2019
NAVSEAINST 8020.6	Department of the Navy Weapon Systems Explosives Safety Review Board	C	11 March 2008

NUMBER	TITLE	REVISION	DATE
NAVSEAINST 8020.8/TB 700-2	Department of Defense Ammunition and Explosives Hazard Classification Procedures	C	30 July 2012
NAVSEAINST 8030.3	Requirements and Procedures for Navy Gun Weapon System Conventional Ammunition Fielding	-	31 January 2019
STANAG 4355/AOP-4355	The Lieske Modified Point Mass and Five Degrees of Freedom Trajectory Models	Edition 4	14 September 2017
STANAG 4547	Design Requirements for Inductive Setting of Medium Calibre Electronic Projectile Fuzes	Edition 2	8 October 2019
STANAG 4624	30 mm x 173 Ammunition	Edition 1	12 September 2012

3.0 Questions

All questions relating to this RFI shall be submitted in writing only via email to the following POCs: Contract Specialist, Kendel Phillips, at kendel.phillips@usmc.mil and Contracting Officer, Benjamin Hunsberger, at Benjamin.j.hunsberger@usmc.mil. The subject line on the email submissions for questions or requests for clarifications should read: RFI: Airburst RFI, Questions/Clarifications. Absolutely no telephone calls will be accepted in response to this RFI. Questions or requests for clarification shall NOT contain proprietary or classified information. To facilitate a timely response, it is requested that all questions or requests for clarification be received by 4:00pm local time Quantico VA on 15 May, 2025. Responses to the questions received will be answered and posted for all interested parties to see via an amendment to the RFI on the System for Award Management (SAM) website. The Government reserves the right to not address questions or requests for clarification should it not be pertinent to the RFI.

For those requiring more technical detail, the Performance Specification (PSPEC) is available upon request. Please contact the Contract Specialist if you would like to receive a copy for your review.

4.0 Response

Contractors possessing the requisite skills, resources, and capabilities necessary to perform the stated requirements, within this document and Annex A, within the timeline described above, are invited to respond to this RFI via a submission of an executive summary on 8-1/2 x 11 inch paper, no more than 15 pages in length, using 12-point font, and map to Annex A sections (I.E. sections 1.1.1, 1.1.2 etc...) . All responses must include the following information:

- Contractor name, address, email, telephone number, point(s) of contact
- Commercial Government Entity Code and Unique Entity ID

- Size of business pursuant to North American Industrial Classification System code 332992
- White paper providing capabilities and relevant past performance

Responses shall be submitted via e-mail to Kendel Phillips at kendel.phillips@usmc.mil, no later than 4:00 PM Eastern Time on 30 May 2025 with the subject line "RFI – Airburst." Telephone inquiries will not be accepted or acknowledged, and no feedback or evaluations will be provided to companies regarding their submissions. Interested parties who consider themselves qualified to deliver the described services are invited to submit a response by the due date.

Any subsequent procurement notifications will be posted on the SAM website at <https://www.sam.gov/>. All interested Contractors are encouraged to register on the SAM website to receive email notifications if and when information related to this acquisition is updated, or documents related to this acquisition are issued and available for download. The Government requests that all interested Contractors ensure that they are listed on the "Interested Vendors List" in SAM. As a rule, the Government will not send notifications directly to interested Contractors, and it is recommended that the interested Contractors periodically check this website for updates.

Proprietary information shall be clearly marked on a page-by-page basis. The Government will not assume that any of the information provided by a respondent is proprietary unless it is conspicuously marked on a page-by-page basis. Moreover, the Government does not assume any duty to contact a respondent and inquire whether any unmarked information submitted in response to this RFI is proprietary.

Classified documents shall not be included in your response.

Do not submit proposals at this time. The Government is only seeking sources/information for market research to determine the availability of capabilities in the described areas. Acknowledgement of receipt will not be made. Availability of any formal solicitation will be announced separately. Submittals will not be returned to the sender. Respondents will not be notified of the results of the evaluation. It is the Contractor's responsibility to check the SAM website at <https://www.sam.gov/> to get additional data.

This RFI does NOT constitute an RFP nor a promise to issue an RFP in the future. This RFI does not commit the USG to contract for any supply or service. Further, the USG is not seeking proposals at this time and will not accept unsolicited proposals. Contractors are advised the USG will not pay for any information or administrative costs incurred in response to this RFI.

The information provided in the RFI is subject to change and is not binding on the USG. The USG has not made a commitment to procure any of the items discussed and the release of this RFI should not be construed as a commitment or as authorization to incur cost for which reimbursement would be required or sought. All submissions become USG property and will not be returned.